

UI/UX REDESIGN PROPOSAL · APPS/WEB

From a basic editor to a *research workspace*.

This proposal covers the dashboard, editor, research and collaboration panels, templates and mobile layout. It adds new functions such as an outline, a submission checklist, a command palette, version history and sharing. Every screen is built on data Colres already stores.

13 pages

Audit · system · screens · roadmap

Stack

Next 16 · Tailwind 4 · shadcn/ui · TipTap · Liveblocks · Convex

September 2026

Design only. No code changed.

UX audit of the current web app

The engine underneath is strong: pagination, citation numbering, APA/IEEE formatting, cross-references and live collaboration all work. The interface doesn't show that strength yet. These are the friction points found in the current code.

AREA	WHAT HAPPENS TODAY	IMPACT ON USERS	SEVERITY
App header AppShell.tsx	A lime-green bar holds only the Clerk buttons on the right. There is no logo, navigation, search or way back home.	No sense of place. Users can't return to documents without the back button, and the colour clashes with the violet accents.	High
Manual save next to live collaboration docs/[editor]/page.tsx	A Save button, a pingin "Unsaved changes" dot and a beforeunload prompt sit on top of a Liveblocks-synced document.	Mixed mental model. Co-authors can't tell what is persisted, and the constant ping adds anxiety.	High
Side panels are overlays	Research and Collaboration open as drawers that cover the page. Only one can be open at a time.	You can't read a paper and cite it at the caret at the same time, which the code comments themselves call "one uninterrupted move".	High
Three stacked sticky bars	Title header, meta bar (Page View checkbox, status, print) and a flat toolbar of about 30 buttons, pinned at top-[128px].	About 190px of chrome before the page. The toolbar wraps on laptops and the tools are hard to scan.	Medium
Template structure unused in the editor	Templates define sections with targetWords, maxWords, required and guidance. None of it is shown while writing.	Authors can't see their progress or submission readiness. A key selling point stays hidden.	Medium
Dashboard docs/page.tsx	A 5xl "👋 Hello" heading. Cards show stripped raw HTML. There is no search, sort, filter, list view, starring or "shared with me". The empty state uses a frowning face.	Doesn't scale past about 9 documents. The raw text preview adds noise, and the empty state reads as negative.	Medium
Status label	doc.status ? 'Draft' : 'Published' on cards, while the editor hard-codes "• Draft".	The labels may be inverted and the two screens disagree. Verify against the Convex mutation.	Medium
Micro-typography and accessibility	Tab labels at 10px, meta text at 10–11px, icon buttons with only title tooltips, and a colour-only save state.	Below WCAG readable minimums. Screen readers and keyboard users get little context.	Low

Design thesis: keep the page central and let everything else sit at the edges. Show the structure the template already knows, and make citing, commenting and checking readiness possible without leaving the document.

Design system: "Ink & Paper"

This is a calm, academic identity. The UI uses a sans-serif and neutral warm greys. The page uses a serif, like a real manuscript. One confident blue is used for actions and amber only for attention. It maps onto the existing shadcn CSS variables in @repo/ui.

Colour tokens



ink
#16203A · foreground

primary
#3B5BDB · actions

primary-soft
#EDF2FF · selection

accent
#F08C00 · attention



canvas
#F6F5F2 · app bg

surface
#FFFFFF · cards, page

border
#E7E4DD

muted-fg
#6B7280

• success #2F9E44 • warning #E67700 • danger #E03131

Collaborator colours come from a fixed eight-hue ring (violet, cyan, orange, green, raspberry and others). Each user keeps the same colour for cursors, avatars and contribution charts. Every text pairing passes WCAG AA at 4.5:1.

Typography

Inter / Geist

UI: 12 · 13 · 14 · 16 · 20 · 24 · 30. Minimum 12px for any label.

Source Serif 4

Document canvas only: 12pt body, 1.6 leading. It gives a manuscript feel and matches academic PDF output.

JetBrains Mono

Citation keys, DOIs, BibTeX and shortcuts.

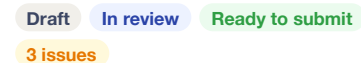
Shape, space & motion

- **Spacing:** 4px base: 4 · 8 · 12 · 16 · 24 · 32 · 48
- **Radius:** 6 for inputs and buttons, 10 for cards, 14 for dialogs, full for chips and avatars
- **Elevation:** 0 flat · 1 card · 2 popover · 3 dialog. The page sheet always sits at level 2.
- **Motion:** 150ms ease-out for hover, 200ms for panels. No bounces or pings. Honour prefers-reduced-motion.
- **Icons:** Lucide at 16px with a 1.75 stroke. Every icon-only button gets an aria-label and a tooltip with its shortcut.
- **Targets:** at least 32px on desktop and 44px on touch

Buttons



Status chips



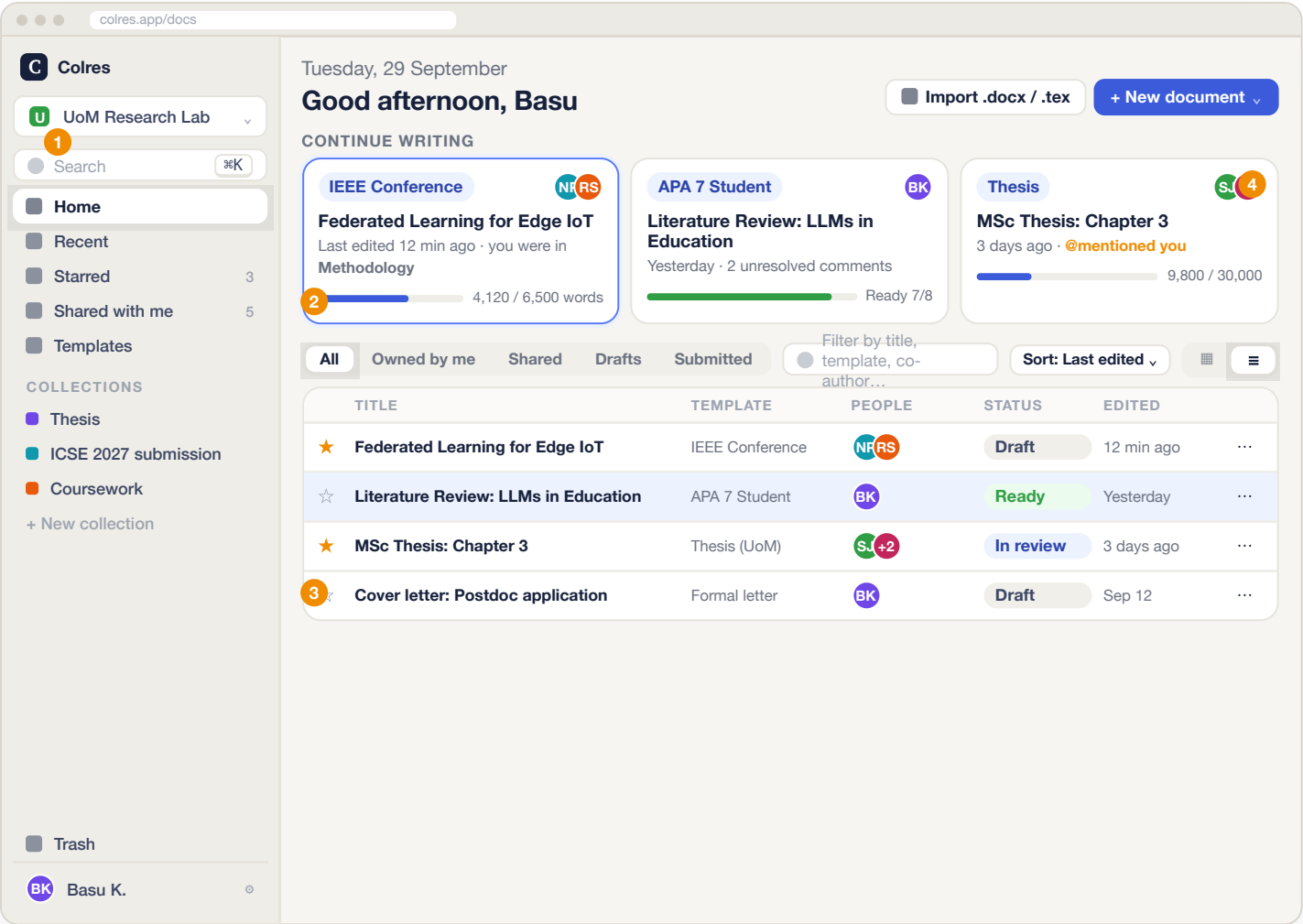
Input + search



Save indicator



Dashboard: an app shell with a document library



1 Persistent sidebar replaces the lime header

It holds the logo (always links home), the org switcher at the top (moved from the header), global search, and the user menu at the bottom. It collapses to a 56px icon rail with `⌵`.

2 "Continue writing" row

The three most recent documents appear with their template, co-authors, the section you last edited, and word progress against the template's targetWords. Mentions are highlighted.

3 A library that scales

Filters (Owned, Shared, Drafts, Submitted), a text filter, sorting, a grid/list toggle that is remembered per user, and starring. Collections add grouping with a collections table.

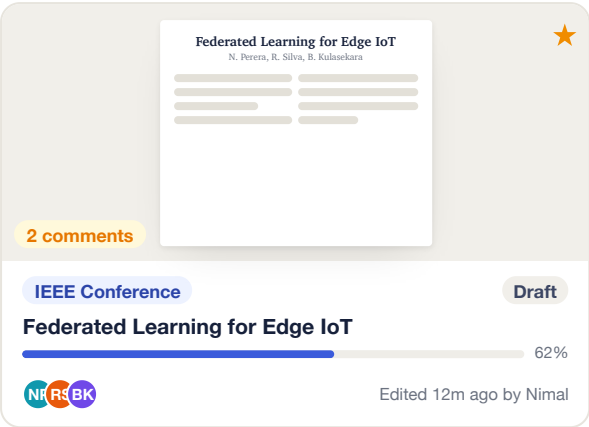
4 Split "New document" button

The main click opens a blank document. The caret shows recent templates and "Browse all templates...". Import brings in .docx or .tex files.

Remove: the 5xl "👋 Hello" heading, the raw HTML text preview on cards, and the `bg-lime-300` header.

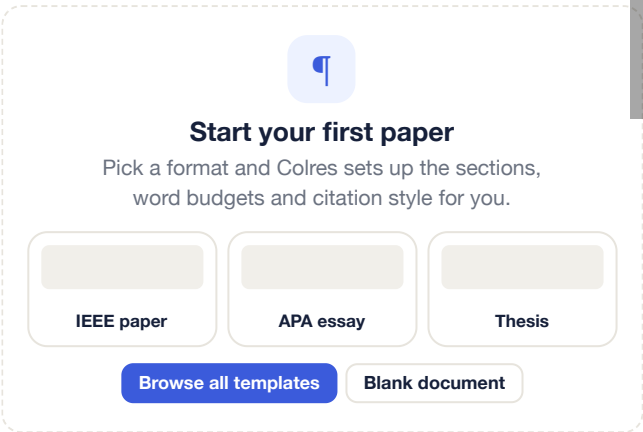
Cards, empty states and safe actions

Document card (grid view)



- The thumbnail is a **first-page render** in the template's layout, not stripped text.
- The whole card is the link. A **⋮ menu** on hover or focus offers Open in new tab, Rename, Duplicate, Star, Move to collection, Download PDF and Delete.
- Keyboard: arrow keys move between cards, Enter opens, **S** stars, **⌫** deletes with undo.

First-run empty state (replaces 🙄)

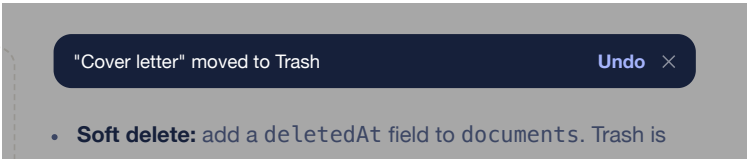


There is a separate, lighter empty state for each filter, for example "No documents shared with you yet. Ask a co-author to invite you by email."

Loading

Keep the existing skeletons but give them the final card shape, so nothing jumps when data arrives. Show the sidebar immediately, because it doesn't depend on Convex.

Delete with undo, not a modal



- **Soft delete:** add a `deletedAt` field to documents. Trash is kept for 30 days and a scheduled Convex job purges it.
- The `AlertDialog` is used only for **permanent delete** from Trash, and for documents with co-authors ("3 people will lose access").

Bulk selection



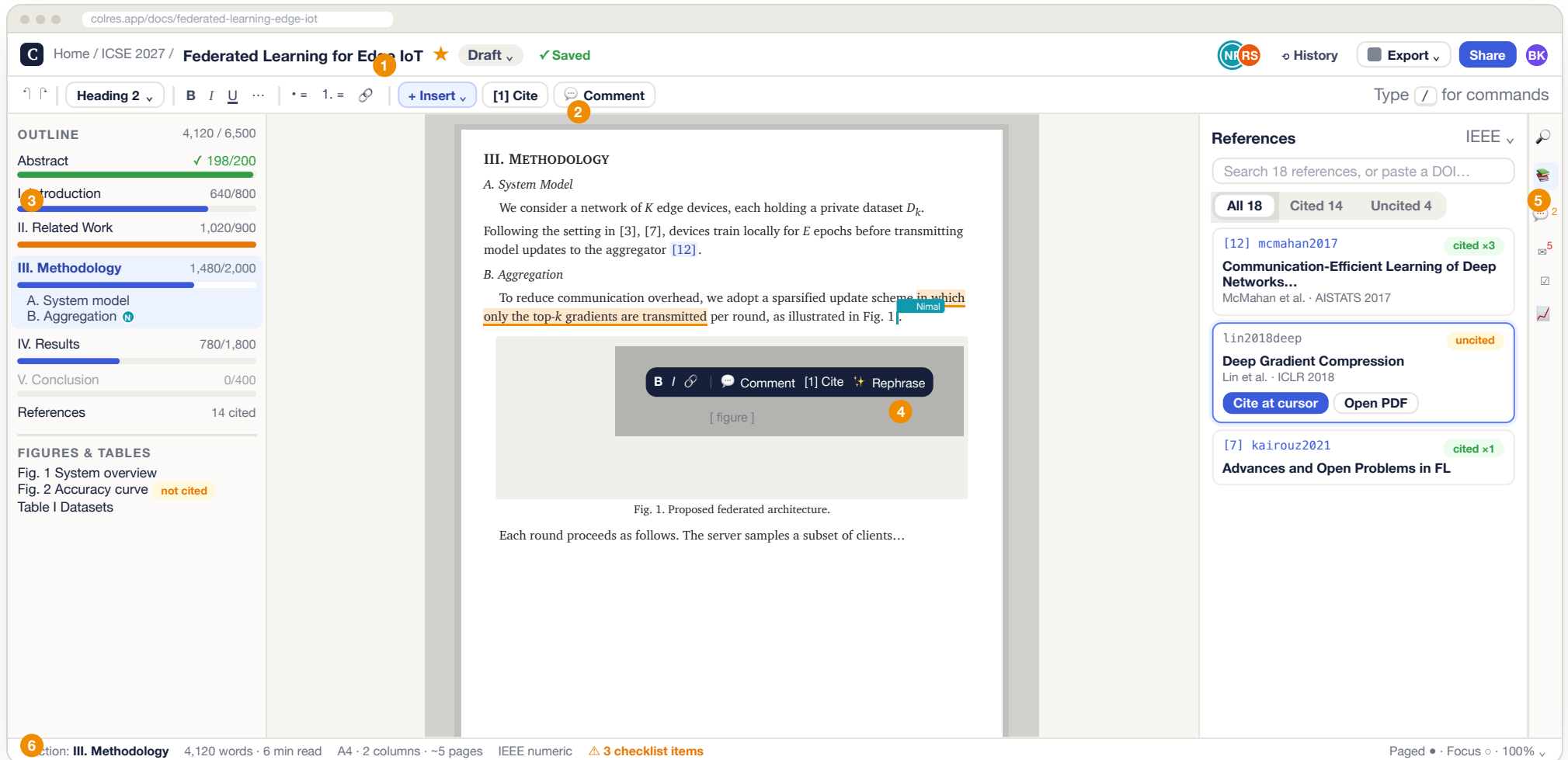
Shift-click or checkboxes in list view select several documents. The action bar replaces the filter row while a selection exists.

Consistent status model



Replace the boolean status with a union. "Ready" is computed from the submission checklist (page 08) and shown the same way on the dashboard and in the editor.

A three-zone writing workspace



1 One slim header. Breadcrumb, editable title, status menu and autosave state replace the Save button.

2 Grouped toolbar. About 12 controls instead of 30. Floats, tables and cross-refs move into *Insert*.

3 Outline + budgets. Shows template sections with live word counts and co-author locations. Click to jump.

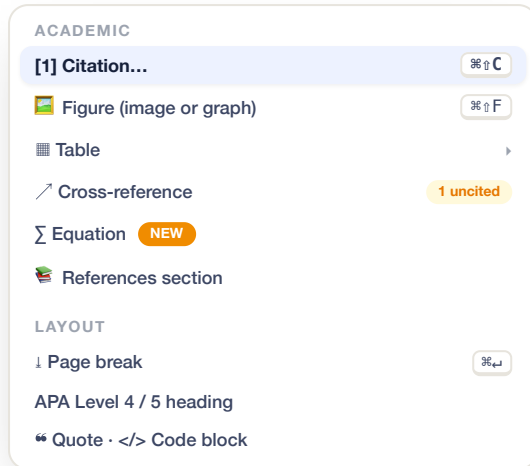
4 Selection toolbar. Comment, Cite and AI rephrase right at the text, extending today's FloatingToolbar.

5 Docked panel + icon rail. It pushes the page instead of covering it. Research, References, Comments, Chat, Checklist and Activity each have a badge.

6 Status bar. Page geometry, count and style move out of the top meta bar. The Page View checkbox becomes a view toggle.

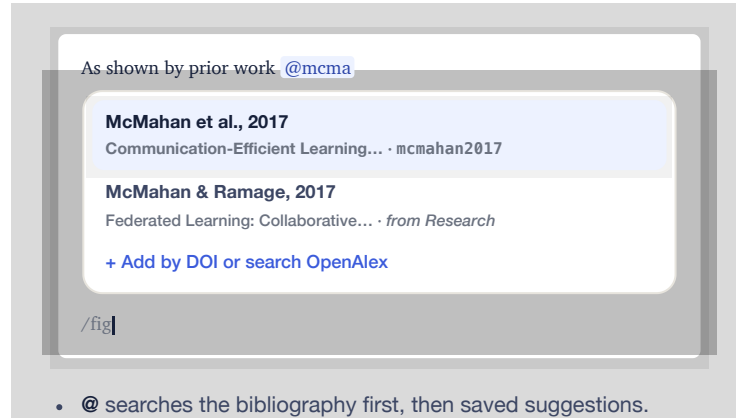
Less chrome, faster insertion, honest saving

Insert menu (groups today's 12+ buttons)



Items adapt to the template. APA-only headings appear only for APA documents, and "Fig. N." captions follow IEEE when the template uses IEEE.

Slash commands & @-cite in the canvas



- @ searches the bibliography first, then saved suggestions. Choosing a suggestion adds it to References and cites it in one step.
- / opens the Insert menu inline. A figure opens the file picker, and so on.
- Markdown shortcuts (##, -, 1.) keep working.

Autosave state machine

✓ **Saved** Idle. The tooltip shows "All changes saved · 14:02".

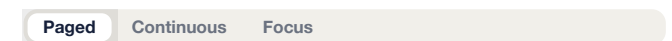
⌚ **Saving...** Debounced 1.5s after typing stops. No ping animation.

⚠ **Offline** Liveblocks is disconnected. Edits are kept locally and a banner explains.

✗ **Couldn't save** After 3 retries. Actions: Retry, Download copy.

- Remove the manual **Save** button. Persist the Liveblocks document to Convex with a debounced mutation or webhook, so the content snapshot stays fresh for thumbnails and search.
- ⌚ becomes "**Save named version**", a checkpoint in History (page 09).
- The beforeunload prompt only appears while "Saving..." or "Offline".

View modes



Focus hides the outline, panel and toolbar, keeps the status bar, and dims paragraphs other than the current one. Toggle with ⌚.

Research, references and a new submission checklist

Research

OpenAlex

Suggestions are based on your title and abstract. Context changed since last run. [Refresh](#)

Suggested 12

Saved 5

Dismissed

Deep Gradient Compression: Reducing the Communication Bandwidth...

Lin, Han, Mao +2 · ICLR 2018 · 2,140 citations · [Open access](#)

✦ Directly supports your top-k sparsification claim in III-B.

[+ Add & cite](#)
[Save](#)
[Abstract ▾](#)
×

Federated Optimization in Heterogeneous Networks

Li, Sahu, Zaheer +3 · MLSys 2020 · 3,900 citations

- "Add & cite" saves the paper to References and inserts it at the caret in one click. The editor stays visible because the panel is docked.
- Show the model's reason line so the suggestion is explained.
- Filters: year range, open access only, minimum citations.

References

IEEE ▾

Paste a DOI, BibTeX, or URL, or drop PDFs here

We detect the format automatically · [Add manually](#)

Sorted by first citation (IEEE)

Group: none ▾

[1] H. B. McMahan *et al.*, "Communication-efficient..." ×3

[2] P. Kairouz *et al.*, "Advances and open..." ×1

— Y. Lin *et al.*, "Deep gradient..." uncited [Cite](#)

! "Edge AI survey" missing year, venue [Fix](#)

- One smart input handles DOI, BibTeX and PDF import (the existing PdfReferenceImport). It detects what was pasted.
- Rows are previewed in the **active style**, so authors see exactly what will print.
- Hovering a row highlights every place it is cited in the document. Clicking cycles through them.
- Validation flags missing fields that the chosen style needs.

Submission checklist

NEW

7/10

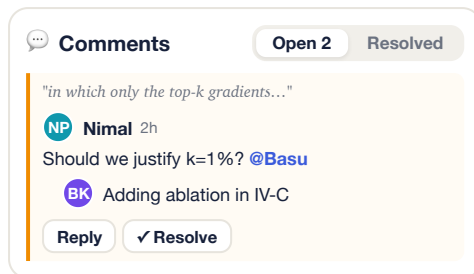
Almost ready for IEEE
 3 items need attention

- ✓ All required sections present
- ✓ Abstract ≤ 200 words
- △ Related Work over budget (1,020/900) [Go](#)
- △ Fig. 2 never referenced in text [Go](#)
- ✗ 1 reference missing year/venue [Fix](#)
- ✓ Every reference is cited
- ✓ Page count ≤ 6 (≈5)
- ✓ No unresolved comments
- Keywords provided

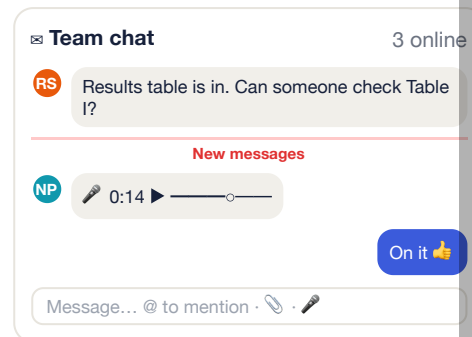
[Export submission PDF](#)

Built entirely from existing data: template sections (required, word limits), FloatNumbering uncited floats, citation order against the bibliography, page count from pageGeometry, the APA running-head limit and open Liveblocks threads. The score feeds the "Ready" status on page 05.

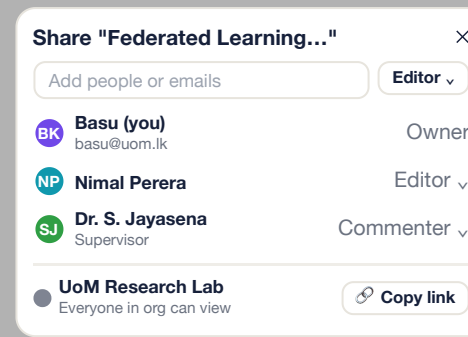
Comments, chat, sharing and version history



- Threads are **anchored to the text**. Clicking scrolls the page and highlights the passage, and highlights in the page open the thread.
- Sort by position in the document, not by time.
- Comments go in a margin gutter when the panel is closed, like Google Docs.



- Keeps today's features (mentions, reactions, voice notes, attachments, unread markers) with better visuals.
- A chat message can **link to a passage**: "III-B" opens that spot.
- Toasts stay, but show at the bottom-left so they don't collide with the panel.



- **Roles:** Owner, Editor, Commenter (suggest mode) and Viewer. These map onto Liveblocks room permissions.
- Invite by email even without a Colres account. The invite resolves on sign-up through Clerk.



- Uses Liveblocks version history for TipTap. Diffs appear inline in the page with coloured additions per author.
- **Contributions** (today's panel) becomes an "Activity" tab next to it, showing the 30-day chart and each member's words, references and comments.

Presence everywhere: avatars in the header show where each person is ("Nimal · Methodology"). Clicking an avatar follows their cursor. The outline shows a small avatar dot on the section each co-author is editing.

Command palette, shortcuts and notifications



⌘K works everywhere. One search covers documents, templates, sections in the current document, references, people and every command. Build it with shadcn Command (cmdk), which is already in the shadcn catalogue used by @repo/ui.

Keyboard shortcuts (open with ?)

⌘K	Command palette	⌘⌘C	Insert citation
⌘S	Save named version	⌘⌘F	Insert figure
⌘\	Toggle sidebar / outline	⌘⌘J	Page break
⌘/	Toggle side panel	⌘⌘M	Comment on selection
⌘.	Focus mode	⌘⌘E	Export
⌘1-6	Switch panel tab	⌘P	Print / PDF preview

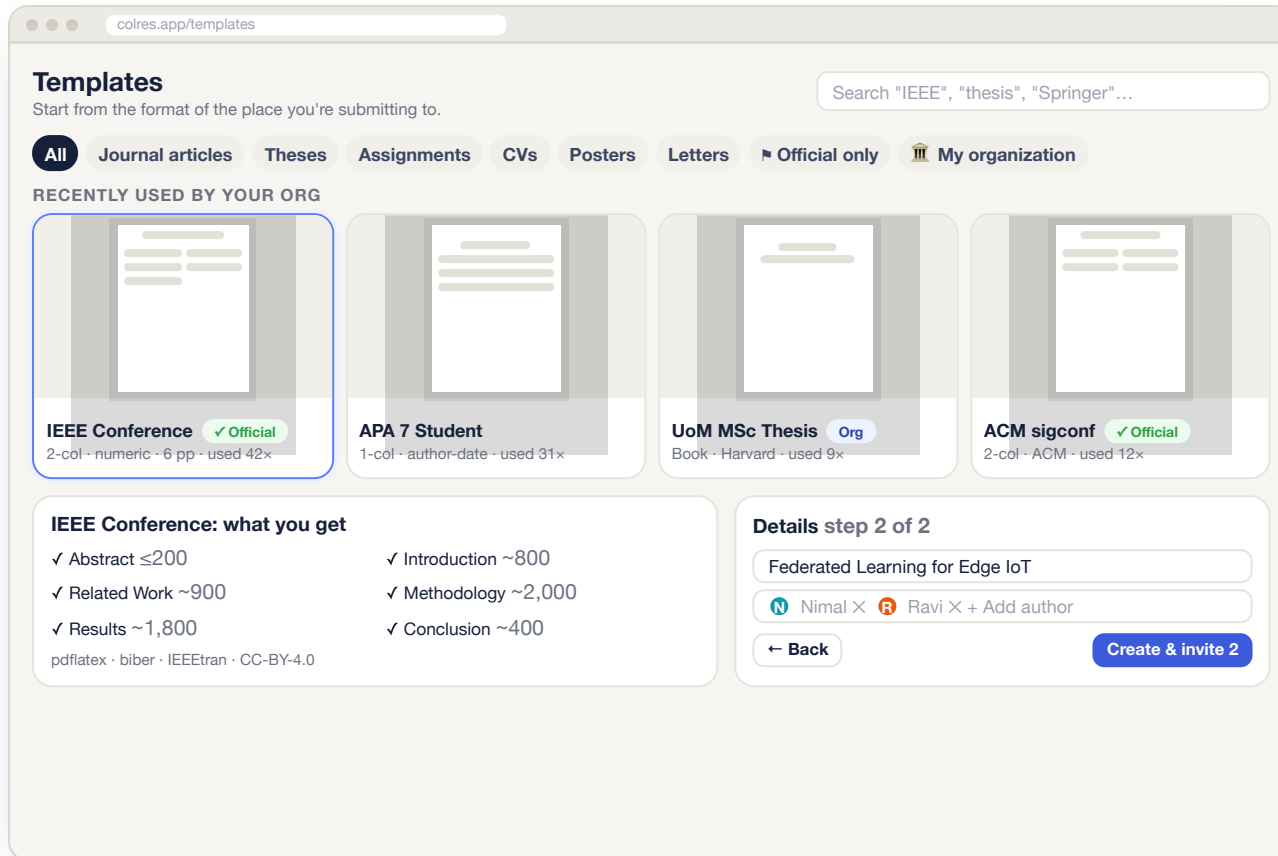
Every toolbar tooltip shows its shortcut, which teaches power use gradually. Windows and Linux show Ctrl.

Notification centre NEW

- NP** Nimal mentioned you in **Federated Learning...** 2h ●
- SJ** Dr. Jayasena resolved 4 comments in **Thesis Ch. 3** 1d
- RS** Ravi shared **ICSE 2027 draft** with you 2d

A bell in the sidebar collects mentions, shares, resolutions and replies across all documents. An email digest is optional and uses Liveblocks notifications.

Choosing a template with confidence



Show outcomes, not metadata

Cards show a real first-page thumbnail (the `thumbnailId` is already in the schema), the layout, the citation style, the page limit and a usage count. Official publisher templates get a trust badge.

"What you get" preview

Selecting a card shows the template's sections with word targets and its compilation contract, so authors know the structure before they commit.

Authors = collaborators

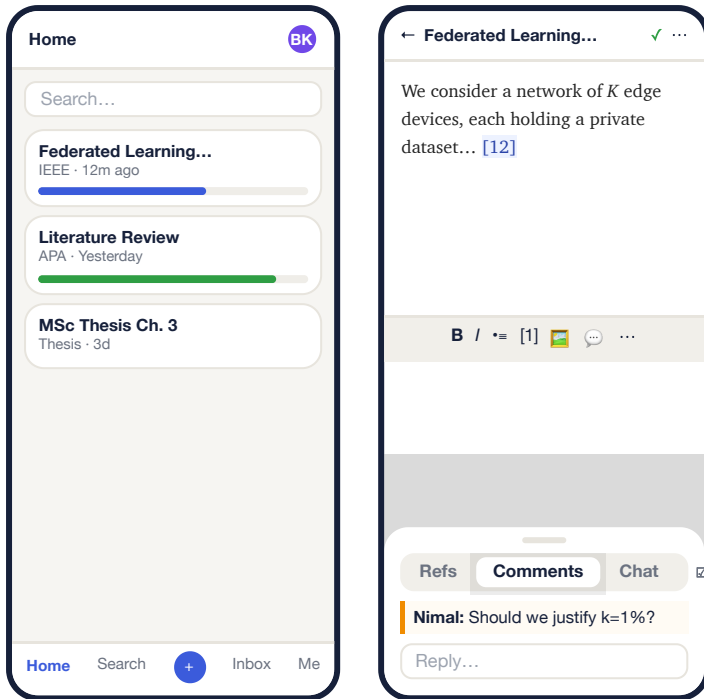
The wizard's authors field autocompletes org members (the `/api/liveblocks-users/search` route already exists). Anyone listed can be invited in the same step.

Two steps, both visible

A progress label ("step 2 of 2") and a live preview of the title page update as fields are filled in. Required fields are marked and checked inline.

Org admins get "Save as org template" from any document's ... menu. It appears under My organization.

Mobile, dark mode and accessibility



Responsive rules

- **≥1280px:** sidebar or outline, page and docked panel all visible.
- **1024–1279px:** the outline collapses to a rail and the panel overlays with a scrim.
- **<768px:** bottom tab bar. The editor becomes continuous (not paged), the format bar sits above the keyboard, and panels become bottom sheets with 3 snap points (the existing vault Drawer).
- Touch targets are at least 44px. Swipe down closes sheets.

Dark mode



The UI chrome goes dark. The **page stays white** by default, since `globals.css` already enforces paper ink. An optional "dark paper" reading mode inverts it for night writing but never affects print.

Accessibility checklist

- WCAG 2.2 AA contrast, with 12px as the minimum text size
- Visible 2px focus ring in `--primary`
- Tabs use Radix Tabs (roles and arrow keys)
- `aria-live` for the save state and chat toasts
- Skip-link to the page. The outline acts as a landmark nav.
- Status is never shown by colour alone (icon plus text)

Implementation roadmap

The work is ordered by user impact per unit of effort. Each phase can ship on its own, and nothing requires rewriting the TipTap extensions.

PHASE 1 · ~1 WEEK

Shell & polish

- Sidebar app shell and removal of the lime header AppShell.tsx, AuthNav.tsx
- Ink & Paper tokens in @repo/ui globals
- Serif document canvas
- Autosave indicator, with the Save button removed docs/[editor]/page.tsx
- Fix the status label and the 10px tab text
- Positive empty state

PHASE 2 · ~2 WEEKS

Editor workspace

- Docked side panel with an icon rail, merging the two Drawers
- Grouped toolbar, Insert menu and status bar TipTap.tsx
- Outline with section word budgets NEW
- Selection toolbar: Cite and Comment
- @-cite and / commands (TipTap Suggestion)

PHASE 3 · ~2 WEEKS

Library & speed

- Dashboard: continue writing, filters, list view and starring docs/page.tsx
- Soft delete, Trash and undo toast
- ⌘K command palette and shortcut sheet
- Template thumbnails and "What you get"
- Submission checklist NEW

PHASE 4 · ~3 WEEKS

Collaboration

- Share dialog and roles NEW
- Version history and compare NEW
- Notification centre NEW
- Collections NEW
- Focus mode, dark mode and the mobile bottom-sheet layout

Schema additions (packages/convex)		
TABLE / FIELD	PURPOSE	USED BY
documents.status → union draft · review · ready · submitted	Clear lifecycle that replaces the boolean	Dashboard chips, editor header, checklist
documents.deletedAt , starredBy[] or a stars table	Trash with undo, and starring	Library, sidebar
collections { orgId, name, colour, documentIds }	User-defined grouping	Sidebar, breadcrumb
documentMembers { documentId, userId email, role }	Per-document sharing and roles, mapped to Liveblocks room access	Share dialog, liveblocks-auth route
versions { documentId, name, snapshotId, authors, createdAt }	Named checkpoints (⌘S) alongside Liveblocks auto history	History panel

How to measure success: time from opening the dashboard to typing (target under 3s), citations added per session, the share of documents reaching "Ready", and fewer panel open/close toggles per session, which shows the docked panel is working.